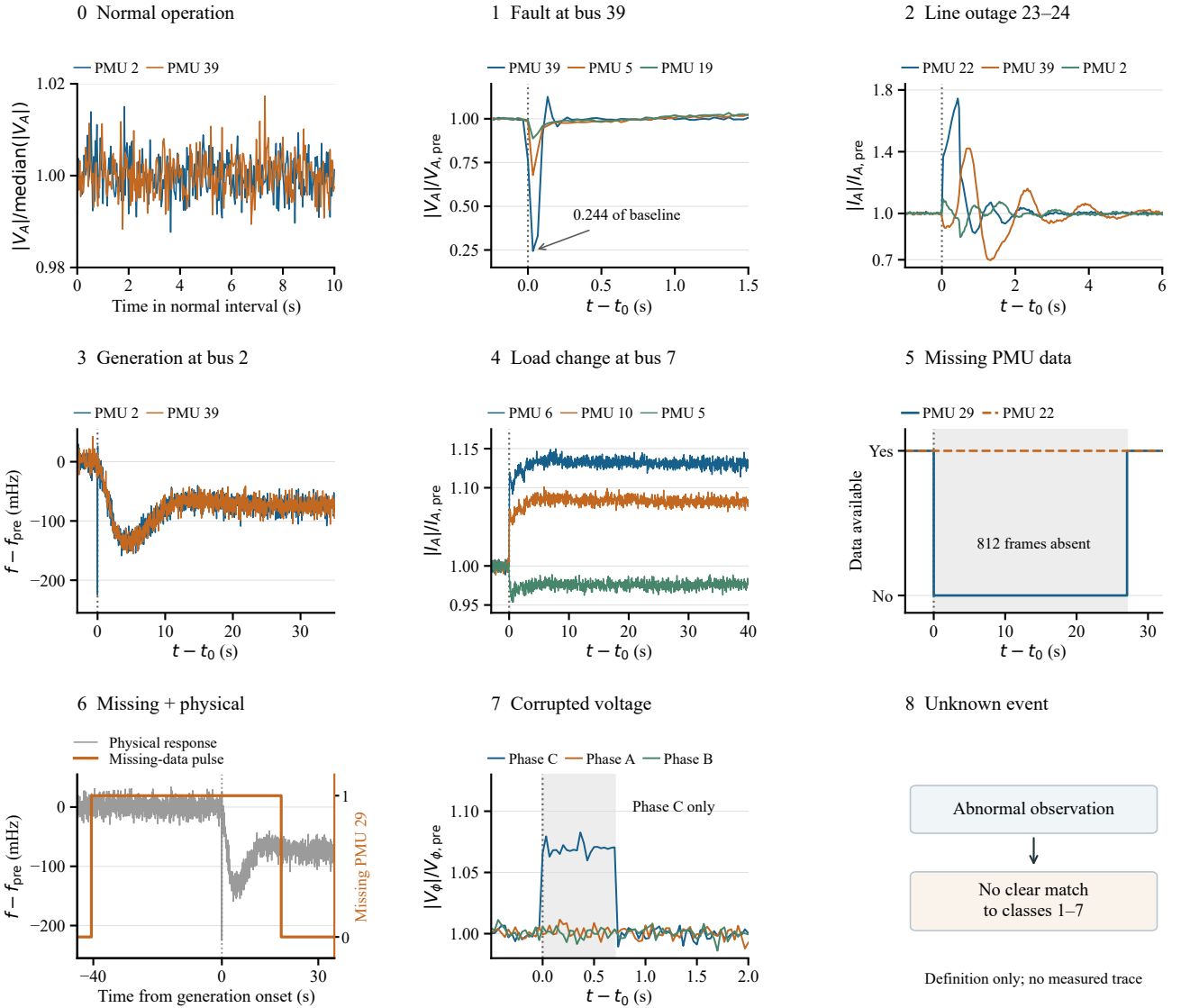

0 Normal operation
PMU 2
PMU 39
$|V_A|/\mathrm{median}(|V_A|)$
Time in normal interval (s)
1 Fault at bus 39
PMU 39
PMU 5
PMU 19
$|V_A|/|V_{A,\mathrm{pre}}$
0.244 of baseline
$t - t_0$ (s)
2 Line outage 23–24
PMU 22
PMU 39
PMU 2
$|I_A|/|I_{A,\mathrm{pre}}$
$t - t_0$ (s)
3 Generation at bus 2
PMU 2
PMU 39
$f - f_{\mathrm{pre}}$ (mHz)
$t - t_0$ (s)
4 Load change at bus 7
PMU 6
PMU 10
PMU 5
$|I_A|/|I_{A,\mathrm{pre}}$
$t - t_0$ (s)
5 Missing PMU data
PMU 29
PMU 22
Yes
No
Data available
812 frames absent
$t - t_0$ (s)
6 Missing + physical
Physical response
Missing-data pulse
$f - f_{\mathrm{pre}}$ (mHz)
Missing PMU 29
Time from generation onset (s)
7 Corrupted voltage
Phase C
Phase A
Phase B
Phase C only
$|V_\phi|/|V_{\phi,\mathrm{pre}}$
$t - t_0$ (s)
8 Unknown event
Abnormal observation
No clear match to classes 1–7
Definition only; no measured trace